CORRELATIONS

Estimation of Entropy of Melting from Molecular Structure: A Non-Group Contribution Method

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The total entropy of melting for a variety of compounds is estimated by a modification of Walden's rule. This modification accounts for the effects of both molecular rotational symmetry and molecular flexibility upon entropy. The simple semiempirical equation gives an average error of 12.5 J/K·mol when applied to more than 930 different compounds.

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Introduction

A knowledge of the entropy of melting is useful for the accurate prediction of chemical properties such as melting point and aqueous solubility (Simamora and Yalkowsky, 1993, 1994; Myrdal et al., 1993; Yalkowsky et al., 1994). Since entropy of melting values are usually not available, an equation for their prediction is needed.

According to Bondi (1968), the total entropy of melting is the sum of its positional, rotational, and conformational components, i.e.:

$$\Delta S_m^{tot} = \Delta S_m^{pos} + \Delta S_m^{rot} + \Delta S_m^{conf} \quad (1)$$

The total entropy of melting can also be defined as the sum of all the entropies of transition and the entropy of melting from the crystal to the isotropic liquid, i.e.:

$$\Delta S_m^{tot} = \Delta S_m + \sum \Delta S_{tr} \quad (2)$$

According to Walden's rule, the entropy of melting for aromatic compounds with little flexibility (e.g., diphenyl and cyclopentane) is constant with a value of 56.5 J/deg·mol. According to Richard's rule, the entropy of melting constant for small spherical compounds (e.g., methane and neon) is 10.5 J/deg·mol. Many compounds do not belong to either of the above categories and, hence, cannot be reasonably estimated.

Recently, Chickos et al. (1990, 1991) developed a general method of predicting the total entropy of melting from group contribution values. Unfortunately, their method is cumbersome and requires the use of numerous group contribution values. Dannenfelser et al. (1993) found that the total entropy of melting for rigid compounds can be predicted quite well by extending Walden's rule to include the effect of molecular rotational symmetry. They related the rotational entropy of melting to the external rotational symmetry number, σ, of the molecule. The rotational symmetry number is a measure of the probability of the molecule being in the proper orientation for incorporation into the crystal (Martin et al., 1979; Dannenfelser et al., 1993). It is related to the rotational entropy by

$$\Delta S_m^{rot} = R \ln \sigma \quad (3)$$

where R is the gas constant. Combining eq 3 and the entropy of melting constant, C, gives:

$$\Delta S_m^{tot} = C - R \ln \sigma \quad (4)$$

Dannenfelser et al. (1993) showed that this equation with a value of 56.5 J/deg·mol for the constant predicts the entropy of melting for rigid molecules quite well.

In this paper the calculation of the entropy of melting is extended and refined so that it is applicable to complex organic nonelectrolytes. The molecular symmetry number, σ, is refined, and the molecular flexibility number, ϕ, is defined. The latter is a measure of the probability of the molecule having the proper conformation for incorporation into the crystal. It is related to conformational entropy by:

$$\Delta S_m^{conf} = R \ln \phi \quad (5)$$

Methods

Molecular Rotational Symmetry Number. A molecule can be oriented in many ways by being rigidly rotated about its center of mass up to 360° in each of the two spherical angles. One of these positions is arbitrarily chosen as the reference orientation. The number of orientations that are identical to the reference orientation is defined as the molecular rotational symmetry number, σ. Examples of symmetry numbers for a variety of molecules are shown in Figure 1. Note that this definition of symmetry differs from that used by crystallographers and that of Domalski and Hearing (1988).

In the assignment of a value to σ, the structure is hydrogen suppressed and the following groups are assumed to be radially symmetrical and/or freely rotating: halogens, methyl, hydroxyl, mercapto, amine, and cyano. All other groups, including nitro and carboxyl, are assumed to be asymmetrical. The molecular symmetry number cannot be less than unity since every molecule has at least one identical orientation (produced by a 360° rotation about any axis). Rings with six or fewer atoms are considered to be rigid and symmetrical. For example, cyclopropane, cyclopentane, and the chair conformation of cyclohexane have symmetries of 6, 10, and 6, respectively. Molecules that are conical (e.g., hydrogen cyanide and chloromethane) or cylindrical (e.g., carbon dioxide and ethane) have one infinite

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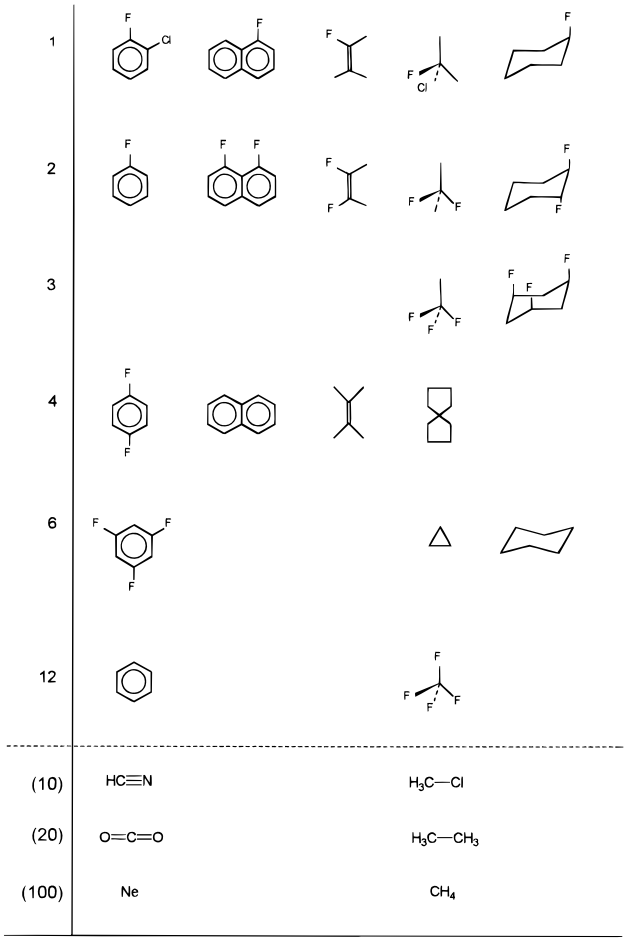


Figure 1. Symmetry numbers for several molecules. Parentheses indicate effective symmetry numbers for molecules with infinite symmetry.

rotational axis and are empirically assigned effective symmetry numbers of 10 and 20, respectively. Spherical molecules (e.g., neon and methane) with an infinite number of infinite rotational axes have been empirically assigned an effective symmetry number 100. (These effective symmetry numbers differ from 20, 20, and 200 for conical, cylindrical, and spherical molecules, respectively, used in a previous report.)

Molecular Flexibility Number. For the simple linear molecules the flexibility number, ϕ, is an exponential function of chain length and can be defined as:

$$\phi = 2.85^{\tau} \tag{6}$$

where τ is the number of torsional angles, and the value 2.85 is based on an equation developed by Temperley (1956) with the assumption that the trans conformation is more stable than the gauche conformation by 2.26 kJ/mol. The number of torsional angles, τ, can be calculated by subtracting 1 from the total number of chain atoms (these do not include end groups, such as the two methyl groups of a linear alkane) present in the molecule. Note that the above-mentioned radially symmetrical end groups, as well as carbonyl oxygen and *tert*-butyl groups, are not included in the number of chain atoms. Hence, for saturated linear molecules τ = SP3 − 1, where SP3 is the number of sp^3 atoms in the chain. For example, 1-chloropropane has a flexibility number of $2.85^{(2-1)}$ or 2.85; similarly, 1-chloropentane has a flexibility number of $2.85^{(4-1)}$ or 23.1, as shown in Table 1. It is assumed that for all sp^3-hybridized atoms the torsional rotation is restricted to a similar extent. Therefore, C, CH, NH, N, O, and S are included in SP3 along with CH_2.

Table 1. Examples of Some Molecular Flexibilities

name	SP3	SP2	RING	τ	ϕ
1-chloropropane	2	0	0	1	2.85
1-chlorobutane	3	0	0	2	8.12
1-chloropentane	4	0	0	3	23.1
2-butanone	1	1	0	0.5	1.69
1-butyne	1	0	0	0	1.0
1-butylnaphthalene	3	0	1	2.5	13.7

For more complex structures sp^2 atoms need to be considered. Since a double bond is less flexible than a single bond, an sp^2 atom contributes less to flexibility than an sp^3 atom. Two sp^2 atoms have been empirically determined to be roughly equivalent in flexibility to one sp^3 atom.

Thus, in calculating the molecular flexibility number, sp^2 and sp^3 chain atoms are assigned values of 0.5 and 1.0, respectively. Since the bonds of sp atoms do not contribute to flexibility, these atoms are given a value of zero. Ring systems, regardless of size, are counted as a single group and are assigned a value of 0.5 per system. A single-ring system consists of a ring or any number of fused rings. Note that the central atoms of nitro, sulfonyl, and carboxyl groups are sp^2 atoms.

From the above, the number of effective torsional angles is calculated for any compound by using the following equation:

$$\tau = \mathrm{SP3} + 0.5\mathrm{SP2} + 0.5\mathrm{RING} - 1 \tag{7}$$

where SP3 is the number of sp^3 chain atoms, SP2 is the number of sp^2 chain atoms, and RING is the number of fused-ring systems. Substituting eq 7 into eq 6 gives:

$$\phi = 2.85^{[\mathrm{SP3}+0.5\mathrm{SP2}+0.5\mathrm{RING}-1]} \tag{8}$$

which is the general equation for calculating the molecular flexibility number for all molecules. It is important to point out here that the exponent is never less than zero.

Examples of some molecular flexibility numbers, ϕ, for complex compounds are shown in the second half of Table 1. The numbers of SP3, SP2, and RING groups are also given.

Data. Evaluated entropy of melting data were obtained from two data compilations by Domalski and co-workers (1984, 1990) and entered into a database using dBASE IV. The database contains 1311 total entropy of melting values which include all of the reported entropies of transition and the entropy of melting for more than 930 different compounds. More than half of these compounds are flexible. Domalski and co-workers assigned ratings to the reported entropy values that range from "A" to "D", indicating "high quality" to "low quality" data. Molecular rotational symmetry and flexibility numbers were assigned to each compound as described above.

Results and Discussion

The result of combining of eqs 4 and 5 contains the effects of both molecular symmetry, σ, and flexibility, ϕ:

$$\Delta S_m = C - R \ln \sigma + R \ln \phi \tag{9}$$

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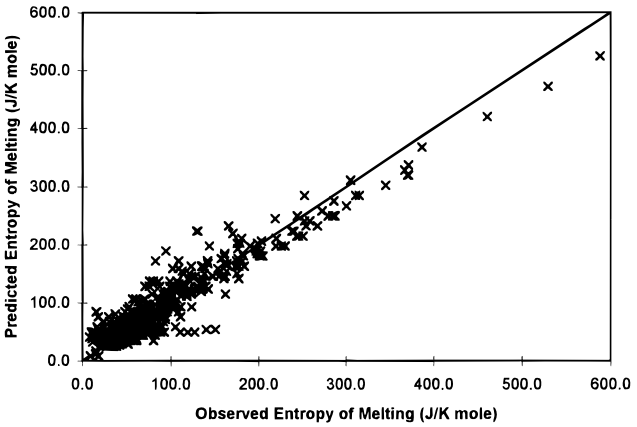


Figure 2. Observed versus predicted entropy of melting for 1277 values: (×) observed data.

The intercept represents those molecules which are rigid and asymmetrical, that is $\ln \phi = 0$ and $\ln \sigma = 0$. The independent data set of Dannenfelser et al. (1993) was used to find the average entropy of melting for rigid, asymmetrical molecules. (Note that the definitions in this paper were used to define the molecular symmetry number.) The average of 245 values was found to be close to 50 J/K·mol. Using this value, eq 9 becomes:

$$\Delta S_m{}^{tot} = 50 - R \ln \sigma + R \ln \phi \text{ J/K·mol} \quad (10)$$

For polymers, the repeating unit values are added to obtain the total entropy of melting, thus, no intercept is used in their calculation.

Thirty-four out of the 1311 values in the database were not used in the evaluation of eq 10 and are indicated in the supporting information table by asterisks. These are grouped into two categories: those that are outliers and those that have missing transitions. The outliers include all reported values of ΔS_m that are clearly out of line with at least two other reported values for the same compound or are less than 9 J/K·mol. The latter are believed to have missing transition data. The data with missing transitions include the values for ethanol which do not report a crystal−crystal transition seen by Nikalaev et al. (1967) and Haida et al. (1977) and the values reported for three discotic liquid crystal forming compounds which are not likely to be in their most stable crystal form except at extremely low temperatures.

The predicted versus observed total entropy of melting values are shown in Figure 2. Perfect prediction is depicted by the solid line. The fact that nearly all of the data, ranging from 9 to 588 J/K·mol, fall close to the line indicates that eq 10 provides a reasonable estimate of the total entropy of melting for a wide variety of compounds. The average absolute error for the 1277 values for 934 compounds is only 12.5 J/K·mol. This is well within the experimental error that is normally associated with observed entropy of melting data.

Table 2 shows the number of entropy values and the absolute error for each of Domalski's ratings. A slightly improved absolute error of 11.0 J/K·mol is achieved when only the 743 high quality or A rated data are used. This is expected since these values should be more reliable than the others.

Table 2. Comparison of Evaluated Data

evaluation	no. of values	average absolute error
A	743	11.1
B	249	15.9
C	244	12.7
D	40	17.1
total	1276	12.5

Conclusion

An equation, based only upon two molecular parameters, σ and ϕ, is developed for the calculation of the total entropy of melting of organic compounds. The equation was tested on an independently reported set of data for 934 compounds and found to provide estimates that are within the range of the experimental error of the measurements.

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Nomenclature

$\Delta S_m{}^{tot}$ = total entropy of melting, J/deg·mol
$\Delta S_m{}^{pos}$ = positional entropy of melting, J/deg·mol
$\Delta S_m{}^{rot}$ = rotational entropy of melting, J/deg·mol
$\Delta S_m{}^{conf}$ = conformational entropy of melting, J/deg·mol
SP2 = sp^2 atoms in a chain
SP3 = sp^3 atoms in a chain
RING = single and fused-ring systems

Greek Symbols

σ = molecular rotational symmetry number
ϕ = molecular flexibility number
τ = effective torsional angles

Supporting Information Available: Data listing the 1311 observed and predicted total entropies of melting for 949 compounds used in this study along with the natural logarithm of molecular rotational symmetry and flexibility numbers, the difference between the observed and predicted values, and Domalski's data evaluation are tabulated. The outliers and the entropy values that have missing transitions are denoted with an asterisk. Both the outliers and the values with missing transitions are not used in the study (34 pages). Ordering information is given on any current masthead page.

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